Matching for a family of symmetric golden maps

Slade Sanderson

with Karma Dajani

Universiteit Utrecht

21 May 2024



Symmetric golden maps

Throughout, $\beta := \frac{\sqrt{5}+1}{2}$ denotes the golden ratio (so $\beta^2 = \beta + 1$).

Define $s : [-1, 1] \to \{0, \pm 1\}$ by $s(x) = i$ iff $x \in J_i$, where
$J_{-1} := [-1, -1/\beta)$, $J_0 := [-1/\beta, 1/\beta]$, $J_1 := (1/\beta, 1]$.

For $\alpha \in [1, \beta]$, define $S_\alpha : [-1, 1] \to [-1, 1]$ by

$$S_\alpha(x) := \beta x - \alpha s(x) = \begin{cases} \beta x + \alpha, & x \in J_{-1}, \\ \beta x, & x \in J_0, \\ \beta x - \alpha, & x \in J_1. \end{cases}$$

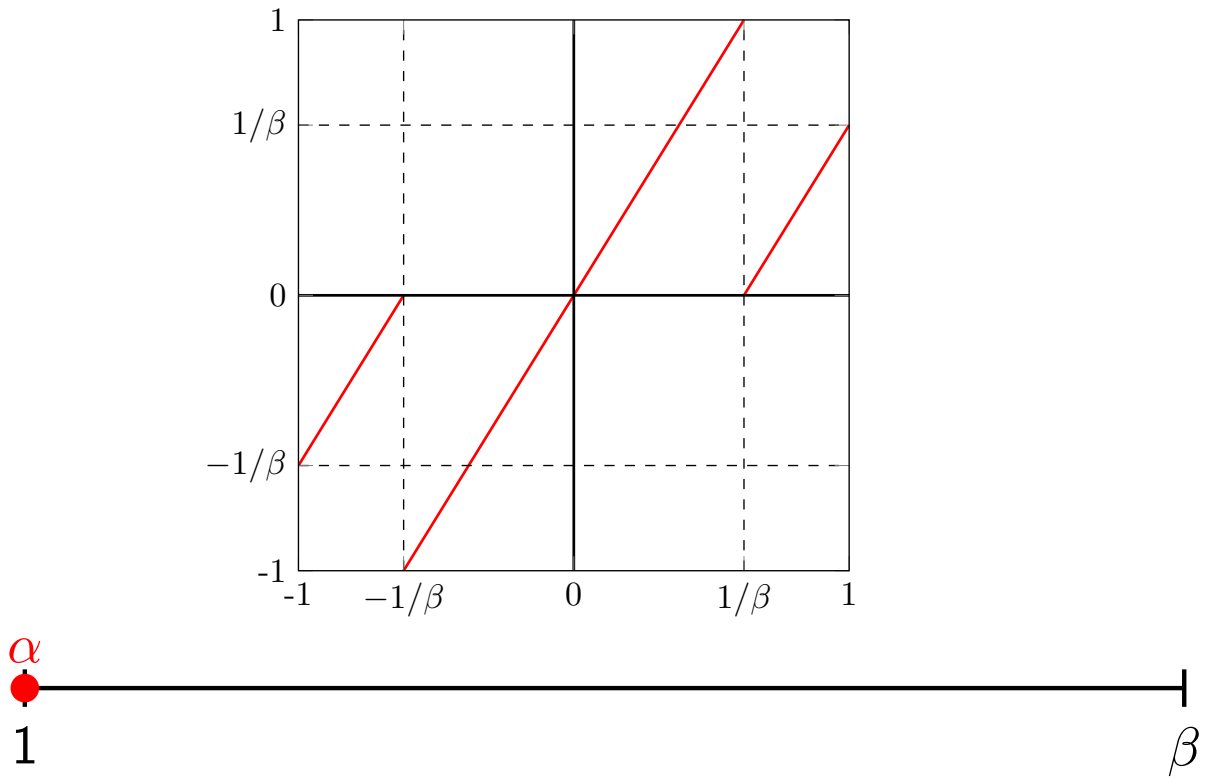


Figure: The map $S_\alpha(x) = \beta x - \alpha s(x)$. Note: $S_\alpha(J_{\pm 1}) \subset J_0$.

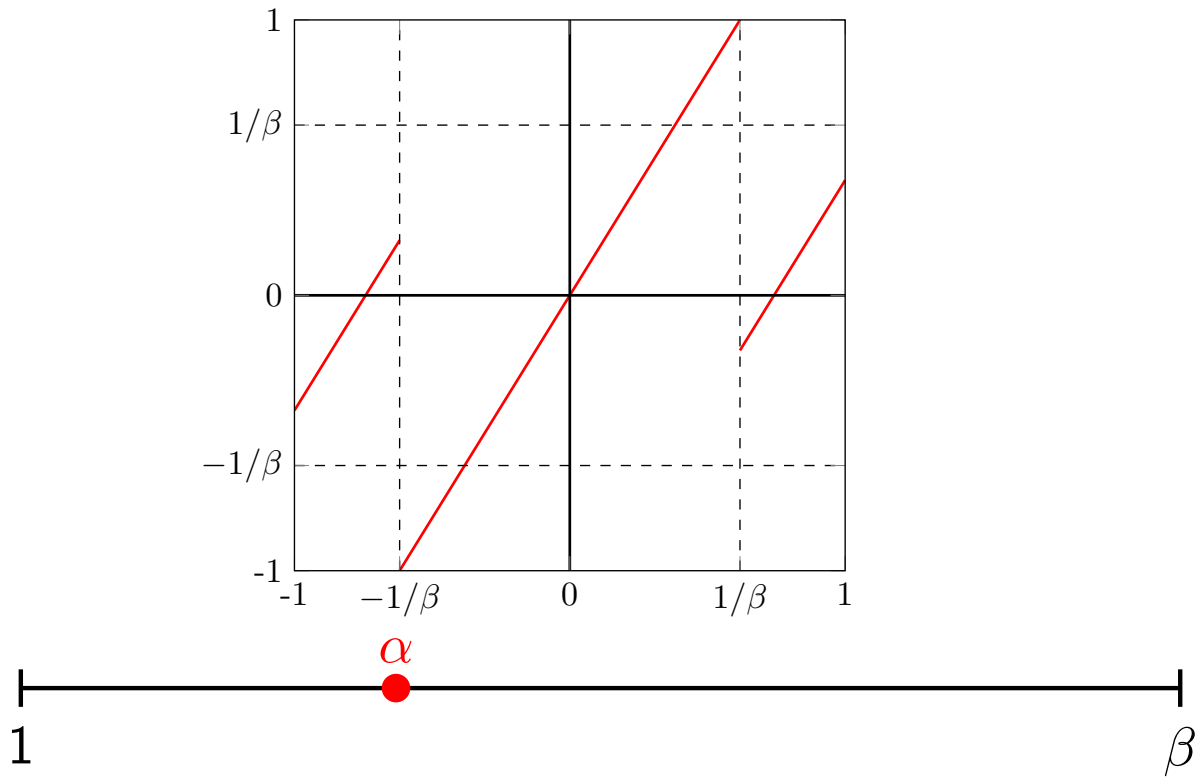


Figure: The map $S_\alpha(x) = \beta x - \alpha s(x)$. Note: $S_\alpha(J_{\pm 1}) \subset J_0$.

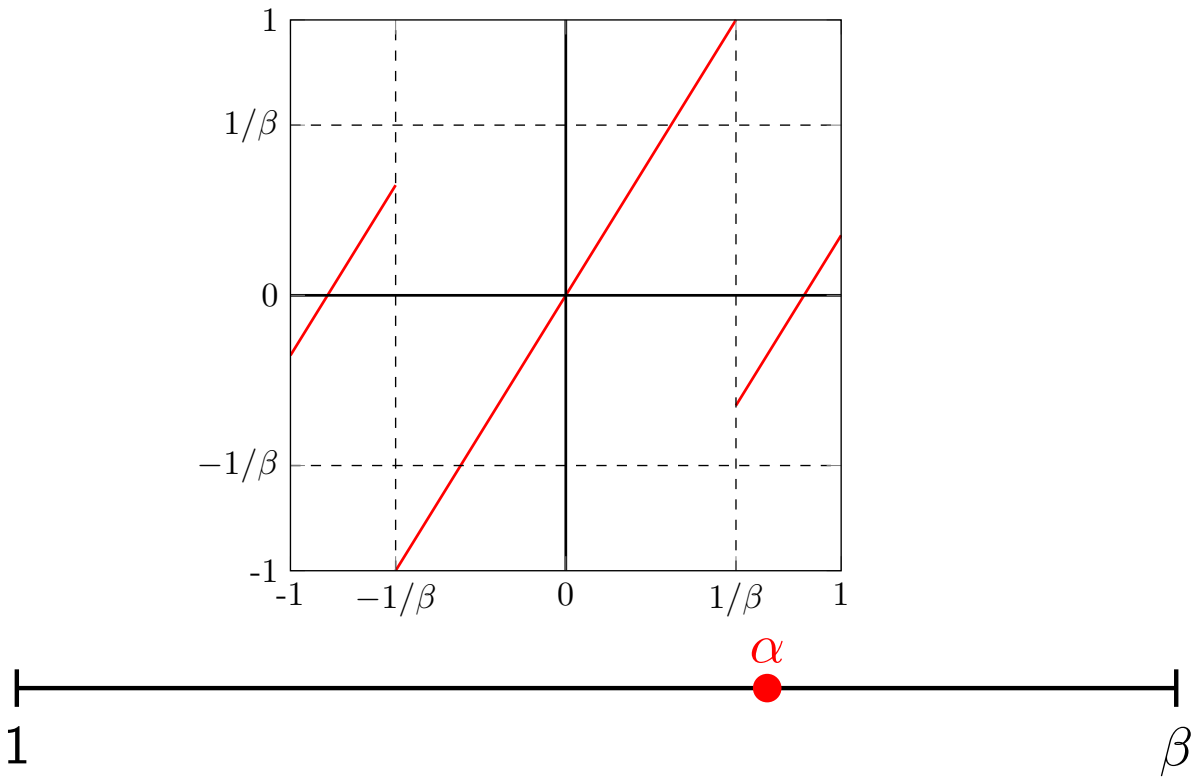


Figure: The map $S_\alpha(x) = \beta x - \alpha s(x)$. Note: $S_\alpha(J_{\pm 1}) \subset J_0$.

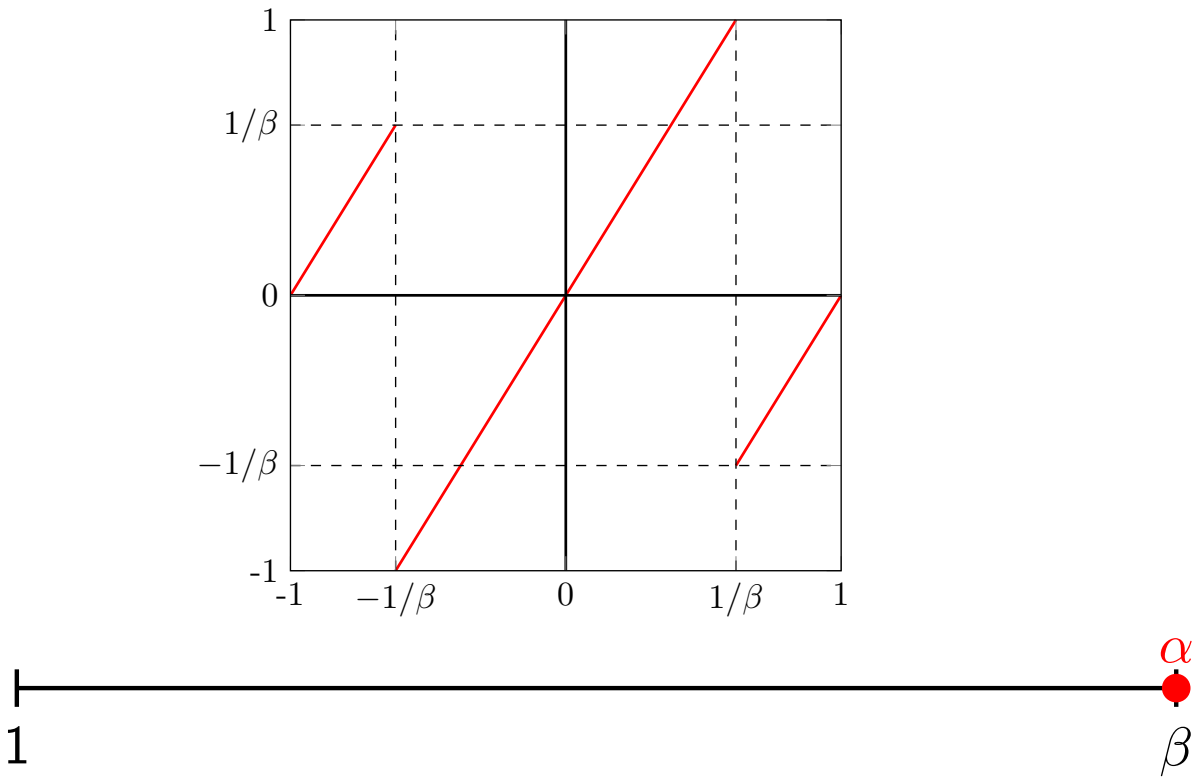


Figure: The map $S_\alpha(x) = \beta x - \alpha s(x)$. Note: $S_\alpha(J_{\pm 1}) \subset J_0$.

S_α-expansions

Recall $S_\alpha(x) = \beta x - \alpha s(x)$. Setting $s_j = s_{\alpha,j}(x) := s(S_\alpha^{j-1}(x))$, $j \geq 1$,

$$S_\alpha^j(x) = \beta S_\alpha^{j-1}(x) - \alpha s_j, \quad \text{or} \quad S_\alpha^{j-1}(x) = \alpha \frac{s_j}{\beta} + \frac{S_\alpha^j(x)}{\beta}.$$

Hence

$$x = \alpha \frac{s_1}{\beta} + \frac{S_\alpha(x)}{\beta} = \alpha \frac{s_1}{\beta} + \alpha \frac{s_2}{\beta^2} + \frac{S_\alpha^2(x)}{\beta}$$
$$= \alpha \frac{s_1}{\beta} + \alpha \frac{s_2}{\beta^2} + \cdots + \alpha \frac{s_j}{\beta^j} + \frac{S_\alpha^j(x)}{\beta^j}.$$

Since $|S_\alpha^j(x)| \leq 1$ and $\beta > 1$, letting $j \to \infty$ gives

$$x = \alpha \sum_{j \geq 1} \frac{s_j}{\beta^j}.$$

Call $(s_j)_{j \geq 1}$ the *S_α-expansion* of x. Q: What 'proportion' of the digits s_j are equal to 0?

Invariant measures, ergodicity and the Ergodic Theorem

Definition

Let μ_α be a measure on $[-1,1]$. We call μ_α *S_α-invariant* if $\mu_\alpha(S_\alpha^{-1}(A)) = \mu_\alpha(A)$ for any $A \in \mathcal{B}$. If, for any $A \in \mathcal{B}$ for which $S_\alpha^{-1}(A) = A$, we have $\mu_\alpha(A) = 0$ or $\mu_\alpha(A^C) = 0$, then μ_α is *ergodic*.

Theorem (Birkhoff, 1931)

If $\mu_\alpha \sim \lambda$ is S_α-invariant and ergodic, then for any $f \in L^1(\mu_\alpha)$,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} f(S_\alpha^j(x)) = \int f d\mu_\alpha \quad \text{for } \lambda\text{-a.e. } x.$$

Setting $f = \chi_{J_0}$ gives the *asymptotic relative frequency of 0*,

$$\lim_{n\to\infty} \frac{1}{n} \#\{1 \le j \le n \mid s_j = s_{\alpha,j}(x) = 0\} = \lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} \chi_{J_0}(S_\alpha^j(x)) = \mu_\alpha(J_0).$$

Main results

Theorem 1

For each $\alpha \in (1, \beta]$, there exists a unique (hence ergodic) a.c. S_α-invariant measure μ_α. In fact, $\mu_\alpha \sim \lambda$. Moreover, there exists a countable collection $\{I_{\mathbf{d}}\}_{\mathbf{d} \in \mathcal{M}}$ of disjoint, open subintervals of $[1, \beta]$ with full λ-measure such that for fixed $\mathbf{d} \in \mathcal{M}$, the density of each μ_α, $\alpha \in I_{\mathbf{d}}$, is a step function with (at most) the same number of jumps.

Theorem 2

The asymptotic relative frequency of 0 of λ-a.e. S_α-expansion is continuous as a function of α and attains its maximum $4/5$ on the maximal interval $[1/2 + 1/\beta, 1 + 1/\beta^2]$.

Kopf's invariant measures

Results of [Kop90] imply that for each $\alpha \in (1, \beta]$, there exists a unique (hence ergodic) a.c. S_α-invariant probability measure μ_α with density

$$f_\alpha(x) = \frac{1}{c} \sum_{j \geq 0} \frac{1}{\beta^{j+1}} \Big(\chi_{[-1, r_j)} - \chi_{[-1, l_j)} - \big(\chi_{[-1, -r_j)} - \chi_{[-1, -l_j)} \big) \Big)(x),$$

where $r_j := S_\alpha^j(1)$, $l_j := S_\alpha^j(1-\alpha)$. Note if $r_j = l_j$ for some $j \geq 0$, then the sum becomes finite and f_α is a step function.

Matching

Definition

We say S_α has *matching* if the *matching index* of S_α,

$$m = m(\alpha) := \inf\{j \geq 0 \mid r_j = l_j\},$$

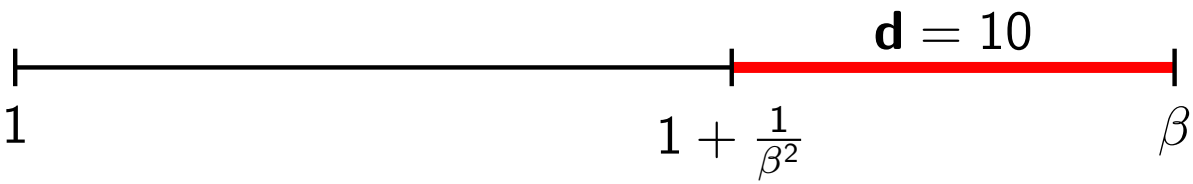
is finite.

Example

Let $\alpha \in (1 + 1/\beta^2, \beta]$. Then

$$r_0 = 1 \in J_1 \qquad l_0 = 1 - \alpha \in J_0$$
$$r_1 = S_\alpha(1) = \beta - \alpha \in J_0 \qquad l_1 = S_\alpha(1-\alpha) = \beta - \beta\alpha \in J_{-1}$$
$$r_2 = S_\alpha^2(1) = \beta^2 - \beta\alpha \qquad l_2 = S_\alpha^2(1-\alpha) = \beta^2 - \beta\alpha$$

Hence $m = 2$, and $\binom{\mathbf{d}}{\mathbf{e}} = \binom{10}{01}$ where $\overline{u} := -u$. Moreover, one verifies that $m = \infty$ for $\alpha = 1 + 1/\beta^2$, so $(1 + 1/\beta^2, \beta]$ is a matching interval corresponding to the matching word $\mathbf{d} = 10$.



Orbit differences

Proposition 1

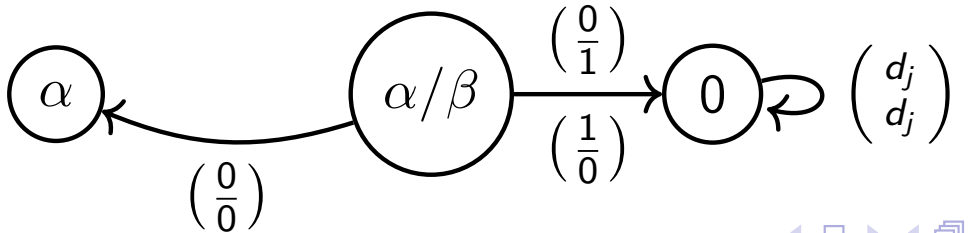
For any $\alpha \in [1, \beta]$ *and* $j \geq 0$, $z_j := r_j - l_j \in \{0, \alpha/\beta, \alpha, \beta\alpha\}$.

Sketch.

Note $z_0 = \alpha$. Suppose $z_{j-1} \in \{0, \alpha/\beta, \alpha, \beta\alpha\}$ for some $j \geq 1$, and note

$$\begin{aligned} z_j &= (\beta r_{j-1} - \alpha d_j) - (\beta l_{j-1} - \alpha e_j) \\ &= \beta z_{j-1} - \alpha(d_j - e_j). \end{aligned}$$

Consider cases: (i) If $z_{j-1} = 0$, then $z_j = 0$. (ii) If $z_{j-1} = \frac{\alpha}{\beta}$, then $\frac{1}{\beta} \leq \frac{\alpha}{\beta} < \frac{2}{\beta}$ implies $\binom{d_j}{e_j} \in \left\{\binom{0}{0}, \binom{1}{0}, \binom{0}{1}\right\}$. (iii) If $z_{j-1} = \ldots$



Orbit differences graph

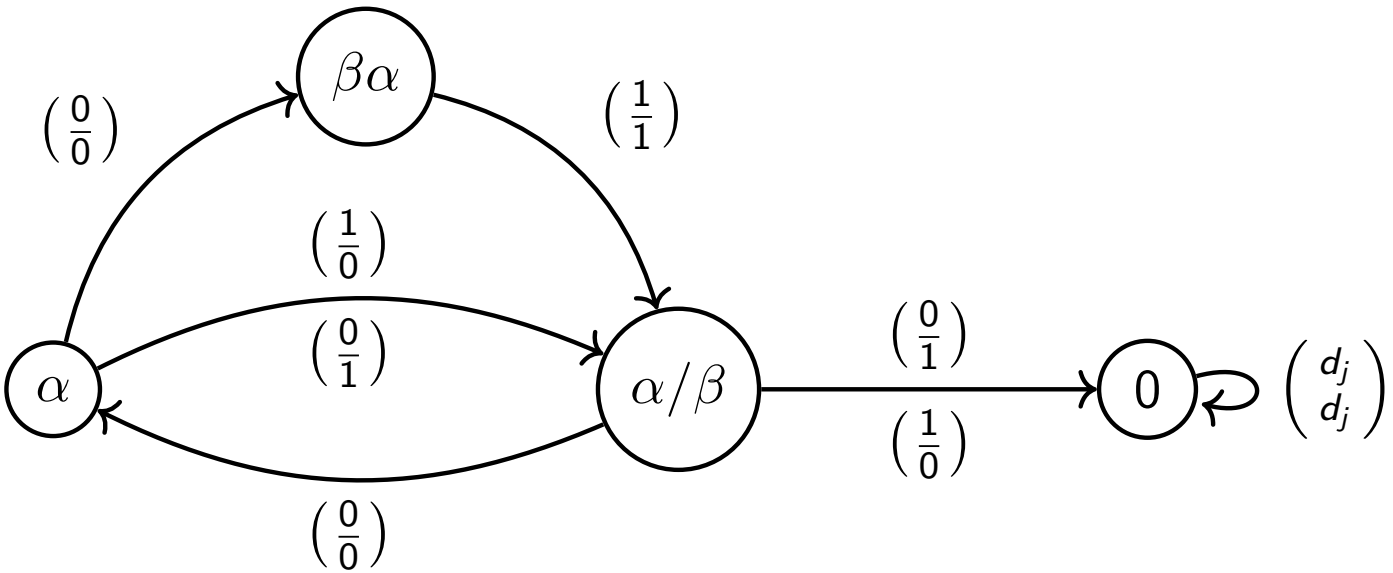
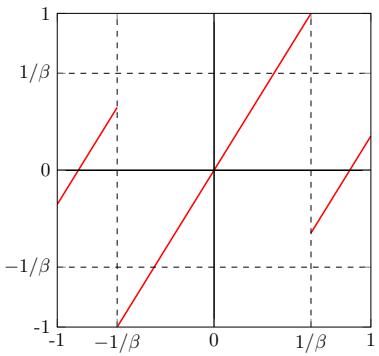


Figure: The *orbit differences graph* $G := (V, E)$, where $V := \{z_{j-1}\}_{j\geq 1} = \{r_{j-1} - l_{j-1}\}_{j\geq 1}$ and $E := \{(z_{j-1}, z_j)\}_{j\geq 1}$ with edge labels $\binom{d_j}{e_j}$.

Matching almost everywhere



Proposition 2

S_α has matching for λ-a.e. $\alpha \in [1, \beta]$.

Sketch.

Notice that $S_\alpha((\frac{1}{\beta}, \frac{\alpha}{\beta})) \subset [-1, 0)$, so if $r_j \in (\frac{1}{\beta}, \frac{\alpha}{\beta})$, there is some $k \geq 0$ such that $r_{j+k} \in J_{-1}$, and the previous Prop. implies S_α has matching. One can relate the beginning of the S_α-orbit $(r_j)_{j \geq 0}$ of 1 to the B-orbit of $1/\alpha$, where $B(x) = \beta x \pmod 1$, and obtain an interval I such that if $B^j(1/\alpha) \in I$, then S_α has matching. Ergodicity of B w.r.t. λ implies this occurs for λ-a.e. α. □

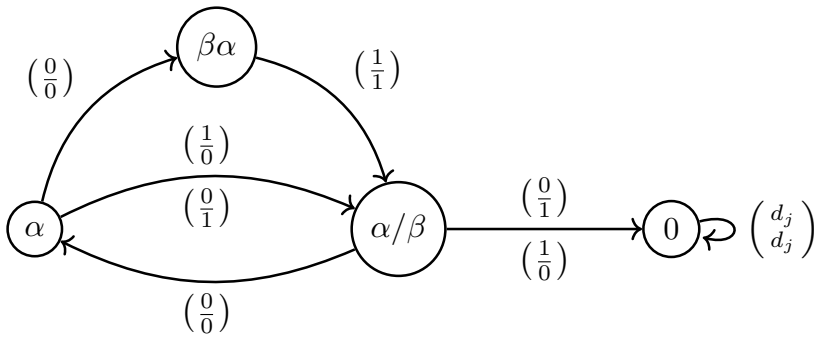
Property M

Let

$$\mathcal{P} := \left\{ \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix} \in (\{0, \pm 1\}^*)^2 \text{ arising from a path in } G \text{ from } \alpha \text{ to } 0 \right\}.$$

Definition

Say $\mathbf{d} \in \{0,1\}^*$ satisfies *Property M* if



(i) $\binom{\mathbf{d}}{\mathbf{e}} \in \mathcal{P}$ for some $\mathbf{e} \in \{0, -1\}^*$,

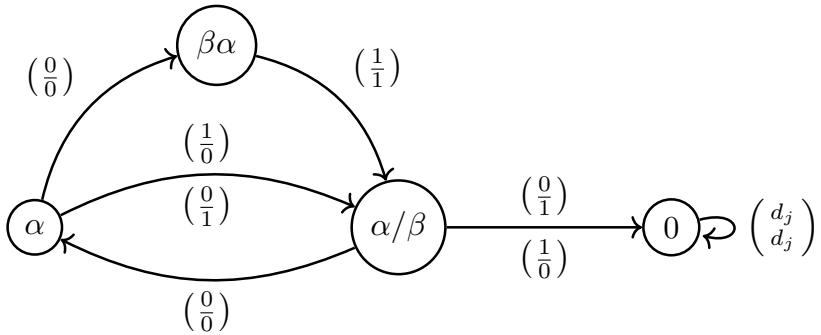
(ii) $d_1 = 1$,

(iii) $d_j d_{j+1} = e_j e_{j+1} = 0$ for all $1 \leq j < |\mathbf{d}|$, and

(iv) $\overline{\mathbf{d}} \preceq \sigma^j(\mathbf{e}) \preceq \sigma^j(\mathbf{d}) \preceq \mathbf{d}$ for all $1 \leq j < |\mathbf{d}|$, with σ the left-shift.

Let $\mathcal{M} := \{\mathbf{d} \text{ satisfying Property M}\}$.

$\mathcal{M}$ contains matching words



Proposition 3

If $\mathbf{d}$ *is a matching word, then* $\mathbf{d} \in \mathcal{M}$.

Proof.

(i) $\binom{\mathbf{d}}{\mathbf{e}} \in \mathcal{P}$: Follows by definition of G and matching.

(ii) $d_1 = 1$: Follows from $1 \in J_1$.

(iii) $d_j d_{j+1} = e_j e_{j+1} = 0$ for all $1 \leq j < |\mathbf{d}|$: Follows from $S_\alpha(J_{\pm 1}) \subset J_0$.

(iv) $\overline{\mathbf{d}} \preceq \sigma^j(\mathbf{e}) \preceq \sigma^j(\mathbf{d}) \preceq \mathbf{d}$: Follows from the facts that $-1 \leq l_j \leq r_j \leq 1$, and $x < y$ iff $(s_{\alpha,j}(x))_{j\geq 1} \prec (s_{\alpha,j}(y))_{j\geq 1}$.



Every $\mathbf{d} \in \mathcal{M}$ is a matching word

Set $I_{10} := (1 + \frac{1}{\beta^2}, \beta]$ and for all $\mathbf{d} \in \mathcal{M} \backslash \{10\}$,

$$I_{\mathbf{d}} := \left(\frac{\beta^m + \beta^{d_m}}{\beta^m v(\mathbf{d}) + \beta^{d_m}}, \frac{\beta^m - \beta^{1-d_m}}{\beta^m v(\mathbf{d}) - \beta^{1-d_m}} \right), \quad \text{where} \quad v(\mathbf{d}) := \sum_{j=1}^{m} \frac{d_j}{\beta^j}.$$

Proposition 4

For every $\mathbf{d} \in \mathcal{M}$, $\varnothing \neq I_{\mathbf{d}} \subset [1, \beta]$, and for any $\alpha \in I_{\mathbf{d}}$, the S_α-expansions of $r_0 = 1,\ l_0 = 1 - \alpha$ begin with $\mathbf{d}$, $\mathbf{e}$, respectively, and $m(\alpha) = |\mathbf{d}|$.

Idea.

For first statement, compare endpoints explicitly. For second statement, again relate beginnings of S_α-orbits of $1,\ 1 - \alpha$ to B-orbits of $1/\alpha,\ 1 - 1/\alpha$. Use B-expansions of $1/\alpha,\ 1 - 1/\alpha$ to show beginnings of S_α-expansions of $1,\ 1 - \alpha$ equal $\mathbf{d}$, $\mathbf{e}$. □

Classification of matching words & intervals

Props. 3 and 4 imply that $\mathcal{M} = \{\text{matching words}\}$. Moreover, Prop. 4 implies every $I_{\mathbf{d}}$ is contained in the matching interval corresponding to $\mathbf{d}$. Conversely,

Proposition 5

If S_α has matching with matching word $\mathbf{d}$, then $\alpha \in I_{\mathbf{d}}$.

Idea.

Use G and the facts that

(i) $I_j = S_\alpha^j(1-\alpha) \in \left(-\frac{\alpha}{\beta}, -\frac{1}{\beta}\right)$ or $r_j = S_\alpha^j(1) \in \left(\frac{1}{\beta}, \frac{\alpha}{\beta}\right)$ for some $j \in \{m-2, m-1\}$, and

(ii) $S_\alpha^j(x) = \beta^j(x - \alpha v(s_1 \dots s_j))$

to obtain bounds on α, which are contained in $I_{\mathbf{d}}$. □

Invariant measures

Recall that for each $\alpha \in (1, \beta]$, there exists a unique (hence ergodic) absolutely continuous S_α-invariant probability measure μ_α with density

$$f_\alpha(x) = \frac{1}{c} \sum_{j \geq 0} \frac{1}{\beta^{j+1}} \Big(\chi_{[-1, r_j)} - \chi_{[-1, l_j)} - \big(\chi_{[-1, -r_j)} - \chi_{[-1, -l_j)} \big) \Big)(x).$$

Theorem 1

For each $\alpha \in (1, \beta]$, there exists a unique absolutely continuous S_α-invariant measure μ_α. In fact, $\mu_\alpha \sim \lambda$. Moreover, there exists a countable collection $\{I_{\mathbf{d}}\}_{\mathbf{d} \in \mathcal{M}}$ of disjoint, open subintervals of $[1, \beta]$ with full λ-measure such that for fixed $\mathbf{d} \in \mathcal{M}$, the density of each μ_α, $\alpha \in I_{\mathbf{d}}$, is a step function with (at most) the same number of jumps.

Frequencies of digits

Let $\alpha \in (1, \beta]$. By Birkhoff's Ergodic Theorem, for a.e. $x \in [-1, 1]$,

$$\lim_{n\to\infty} \frac{1}{n} \#\{1 \le j \le n \mid s_j = s_{\alpha,j}(x) = 0\} = \mu_\alpha(J_0).$$

Define $\mathfrak{f} : (1, \beta] \to [0, 1]$ by $\mathfrak{f}(\alpha) = \mu_\alpha(J_0)$. Arguments analogous to [DK20] imply $\mathfrak{f}(\cdot)$ is continuous. Since matching α are dense, it suffices to study $\mathfrak{f}|_{I_{\mathbf{d}}}$. One finds for all $\alpha \in I_{\mathbf{d}}$,

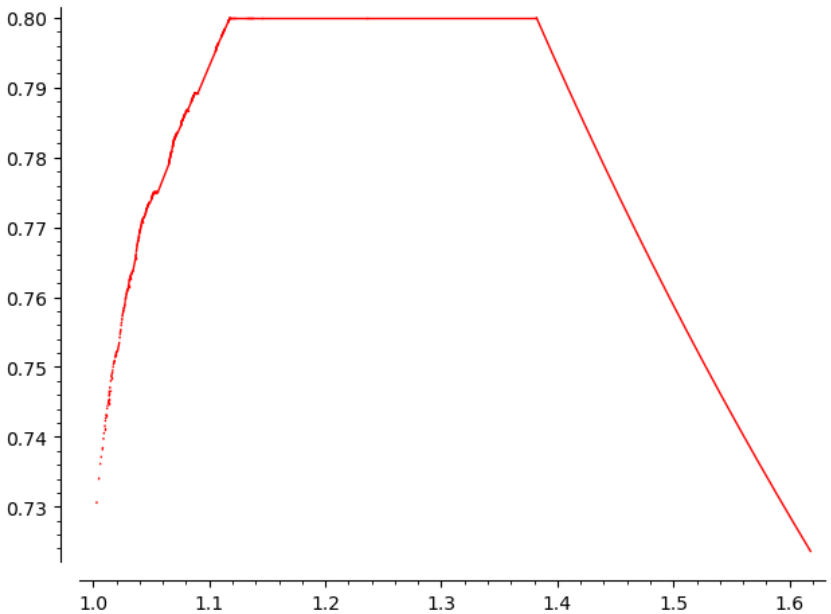
$$\mathfrak{f}(\alpha) = 1 - \frac{1}{\beta v(\Xi(\mathbf{d}))} \left(\frac{n(\mathbf{d}) - 1}{\alpha} - K(\mathbf{d}) \right),$$

where $\Xi(\cdot)$, $n(\cdot)$, $K(\cdot)$ depend only on $\mathbf{d}$.

An analysis of $\mathfrak{f}|_{I_d}$ leads to:

Theorem 2

The asymptotic relative frequency of 0 *of* λ*-a.e.* S_α*-expansion is continuous as a function of* α *and attains its maximum* $4/5$ *on the maximal interval* $[1/2 + 1/\beta, 1 + 1/\beta^2]$.

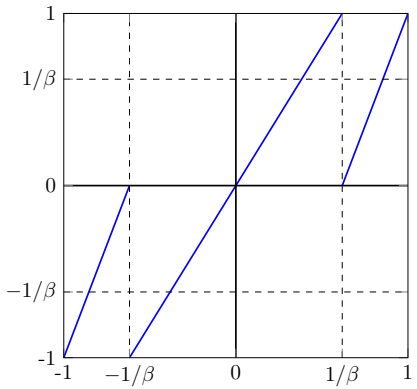


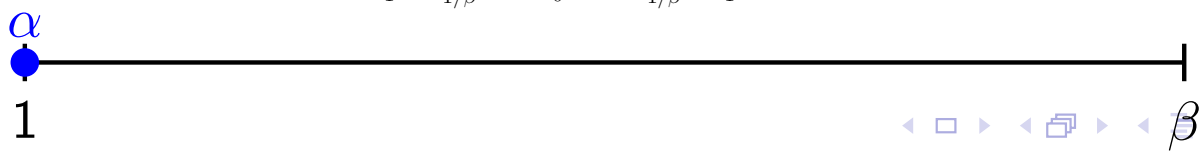
Jump transformations

For $\alpha \in [1, \beta]$, define $T_\alpha : [-1, 1] \to [-1, 1]$ by

$$T_\alpha(x) := \beta^{1+|s(x)|}x - s(x)\beta\alpha = \begin{cases} \beta^2 x + \beta\alpha, & x \in J_{-1}, \\ \beta x, & x \in J_0, \\ \beta^2 x - \beta\alpha, & x \in J_1, \end{cases}$$

i.e. T_α is the *jump transformation* of S_α w.r.t. J_0. Analogous to S_α, each T_α generates an expansion $(t_j)_{j\geq 1} = (t_{\alpha,j}(x))_{j\geq 1}$ of each $x \in [-1, 1]$.



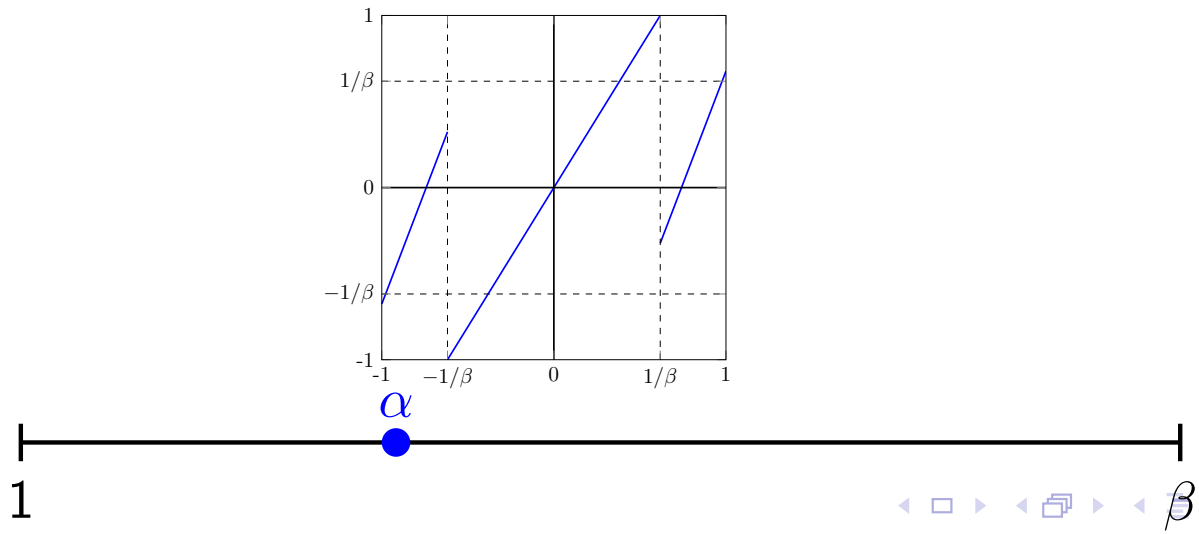


Jump transformations

For $\alpha \in [1, \beta]$, define $T_\alpha : [-1, 1] \to [-1, 1]$ by

$$T_\alpha(x) := \beta^{1+|s(x)|}x - s(x)\beta\alpha = \begin{cases} \beta^2 x + \beta\alpha, & x \in J_{-1}, \\ \beta x, & x \in J_0, \\ \beta^2 x - \beta\alpha, & x \in J_1, \end{cases}$$

i.e. T_α is the *jump transformation* of S_α w.r.t. J_0. Analogous to S_α, each T_α generates an expansion $(t_j)_{j\geq 1} = (t_{\alpha,j}(x))_{j\geq 1}$ of each $x \in [-1, 1]$.

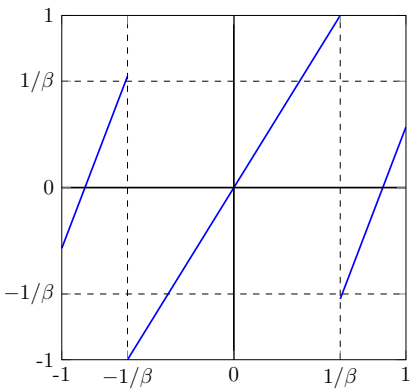


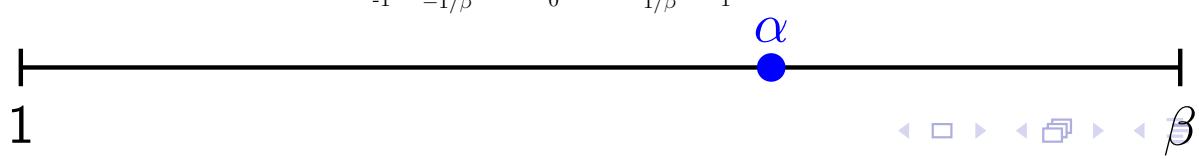
Jump transformations

For $\alpha \in [1, \beta]$, define $T_\alpha : [-1, 1] \to [-1, 1]$ by

$$T_\alpha(x) := \beta^{1+|s(x)|}x - s(x)\beta\alpha = \begin{cases} \beta^2 x + \beta\alpha, & x \in J_{-1}, \\ \beta x, & x \in J_0, \\ \beta^2 x - \beta\alpha, & x \in J_1, \end{cases}$$

i.e. T_α is the *jump transformation* of S_α w.r.t. J_0. Analogous to S_α, each T_α generates an expansion $(t_j)_{j\geq 1} = (t_{\alpha,j}(x))_{j\geq 1}$ of each $x \in [-1, 1]$.



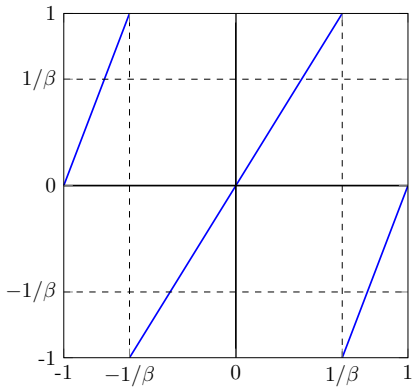


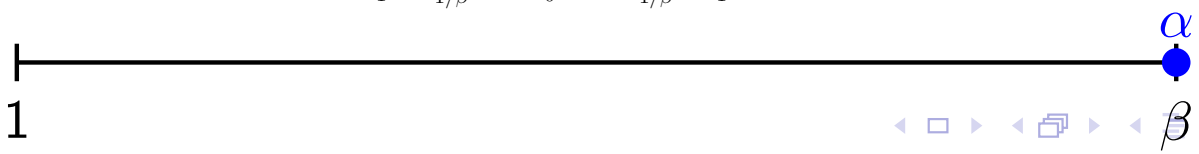
Jump transformations

For $\alpha \in [1, \beta]$, define $T_\alpha : [-1, 1] \to [-1, 1]$ by

$$T_\alpha(x) := \beta^{1+|s(x)|}x - s(x)\beta\alpha = \begin{cases} \beta^2 x + \beta\alpha, & x \in J_{-1}, \\ \beta x, & x \in J_0, \\ \beta^2 x - \beta\alpha, & x \in J_1, \end{cases}$$

i.e. T_α is the *jump transformation* of S_α w.r.t. J_0. Analogous to S_α, each T_α generates an expansion $(t_j)_{j\geq 1} = (t_{\alpha,j}(x))_{j\geq 1}$ of each $x \in [-1, 1]$.





Analogous results

Theorem

For each $\alpha \in (1, \beta]$, there exists a unique absolutely continuous T_α-invariant measure μ_α. In fact, $\mu_\alpha \sim \lambda$. Moreover, there exists a countable collection $\{I_{\mathbf{d}}\}_{\mathbf{d} \in \mathcal{M}}$ of disjoint, open subintervals of $[1, \beta]$ with full λ-measure such that for fixed $\mathbf{d} \in \mathcal{M}$, the density of each μ_α, $\alpha \in I_{\mathbf{d}}$, is a step function with (at most) the same number of jumps.

Theorem

The asymptotic relative frequency of 0 of λ-a.e. T_α-expansion is continuous as a function of α and attains its maximum $3/4$ on the maximal interval $[1/2 + 1/\beta, 1 + 1/\beta^2]$.

References

K. Dajani and C. Kalle, "Invariant measures, matching and the frequency of 0 for signed binary expansions", *Publ. Res. Inst. Math. Sci.*, vol. 56, no. 4, pp. 701–742, 2020.

C. Kopf, "Invariant measures for piecewise linear transformations of the interval", *Appl. Math. Comput.*, vol. 39, no. 2, pp. 123–144, 1990.